

Progressivity and Prosperity: An Econometric Analysis of redistributive spending Structures and inequality.

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May 2026

Abstract

This research investigates the effectiveness of fiscal redistribution in mitigating poverty across OECD nations, specifically examining whether social spending provides sustainable pathways to prosperity or merely temporary relief. Utilizing a panel dataset of OECD member states from 2000 to 2022, the study employs a two-way fixed effects (TWFE) regression model to analyze the relationship between public social spending and poverty rates. Recognizing that policy impacts are rarely instantaneous, social spending is modeled with a three-year lag. Furthermore, the analysis tests the "leaky bucket" hypothesis by incorporating an interaction term between logged social spending and categorical institutional corruption indices. The findings reveal that while social spending has a statistically significant negative effect on poverty ($p < 0.01$), its efficacy is heavily moderated by institutional quality. In low-corruption environments, the poverty-reducing impact of social spending is nearly triple that of high-corruption environments. These results underscore that the mere volume of fiscal redistribution is insufficient for poverty alleviation; rather, the integrity of the state apparatus is a critical determinant of redistributive efficiency.

Introduction

The relationship between fiscal policy and income distribution remains one of the most contentious debates in modern macroeconomics. Central to this discourse is the question of whether public social spending effectively reduces the proportion of individuals living below the poverty line or if it merely provides a short-term safety net without creating sustainable pathways to economic mobility. In an era of increasing

fiscal scrutiny, understanding the conditions under which welfare spending successfully mitigates inequality is paramount for policymakers.

Research Question

The primary objective of this study is to answer the following: *Does a more progressive and redistributive spending policy lead to higher income equality outcomes, and to what extent do institutional frameworks influence this relationship?* Specifically, this paper seeks to quantify how institutional corruption acts as a moderator, potentially diluting the impact of social transfers.

Existing Literature and Contribution

Existing literature, such as Acemoglu et al. (2015), establishes that while democratic transitions expand a government's capacity for redistribution, the direct impact on inequality is inconsistent. This study builds upon that foundation by arguing that the missing link is institutional quality. While traditional economic models often treat the state as a neutral transfer machine, this paper aligns with Okun's (1975) Leaky Bucket hypothesis, which posits that the process of moving money from the rich to the poor involves administrative and institutional leaks.

The contribution of this paper is twofold. First, it moves beyond the simple correlation between spending and poverty by introducing a three-year lag structure, acknowledging the delayed transmission of social policy. Second, it provides empirical weight to the idea of an institutional multiplier, demonstrating that the same dollar of social spending

yields vastly different outcomes depending on the corruption level of the recipient country.

Data and Methodology

To test these dynamics, I utilize a comprehensive panel dataset covering OECD countries from 2000 to 2022. Data for the dependent variable, the poverty rate (defined as the percentage of the population with a disposable income below 50% of the median disposable income), and the primary independent variable, logged public social spending, were sourced from the OECD Social Expenditure Database. To measure institutional quality, I integrated corruption indices from the Global Data Lab (GDL), which were transformed into a categorical variable representing low, medium, and high degrees of corruption.

The econometric strategy employs a Two-Way Fixed Effects (TWFE) model, expressed as follows:

$$\text{Poverty}_{it} = \beta_0 + \beta_1 \text{SocialSpending}_{it-3} + \beta_2 \text{Corruption}_{it} + \beta_3 (\text{SocialSpending}_{it-3} \times \text{Corruption}_{it}) + \beta_4 X_{it} + \alpha_i + \gamma_t + \mu_{it}$$

where X represents a vector of controls including GDP, population size, educational attainment, and unemployment rates. By including country fixed effects, α , and year fixed effects, γ , the model controls for time-invariant country characteristics and global economic shocks, respectively.

Main Findings

The empirical results provide robust evidence that social spending is a powerful tool for poverty reduction, but its success is not universal. The baseline results show a statistically significant negative coefficient for social spending ($p < 0.01$). However, the interaction term reveals a stark disparity: in countries with low corruption, the impact of social spending on poverty reduction is approximately three times more effective than in countries with high corruption levels. This confirms that corruption does not just lead to the capture of funds by elites, but fundamentally erodes the efficiency of the redistributive mechanism.

In conclusion, this paper suggests that for social spending to act as a multiplier rather than just a pathway, fiscal expansion must be accompanied by institutional reform.

Strengthening the state apparatus is as essential to poverty reduction as the funding of the welfare programs themselves.

literature review:

Okun's (1975) work provides a foundational theoretical lens for assessing the efficiency of state intervention, specifically through his characterization of the "institutional framework" as a mediator of economic outcomes. Okun posits that economic disturbances, such as inflation, impose significant welfare costs by creating haphazard redistributions and disrupting institutions that traditionally economize on information and transaction costs. This analysis forms the bedrock of the leaky bucket hypothesis, suggesting that the process of moving resources from one segment of society to another is inherently subject to leaks caused by administrative friction and institutional weaknesses. In the context of poverty alleviation, Okun's framework implies that the

mere volume of fiscal redistribution is insufficient to guarantee equity; rather, the redistributive power of social spending is fundamentally limited by the integrity of the state's administrative bucket. Consequently, where institutional quality is low, such as in high-corruption environments, the leaks in the redistributive process become so pronounced that they can negate the intended impact of even the most progressive fiscal policies.

Acemoglu et al. (2015) explain that while democracy is a robust predictor of increased tax revenue as a fraction of GDP, its direct impact on reducing inequality is less consistent. Their findings suggest that the transition to democracy primarily expands the government's capacity for redistribution by increasing the revenue available for public programs. However, they argue that the actual reduction of inequality depends on whether the political process is "captured" by elites or if it successfully directs those revenues toward broader social outcomes, such as secondary schooling. This implies that the strength of democratic institutions is a necessary precursor to the social spending required to lift low-wage workers above the 25% median threshold.

While government policy sets the stage for redistribution, the baseline level of inequality is often dictated by market frictions. Helpman et al. (2010) argue that structural factors, particularly labor market frictions and differences in firm productivity, are primary drivers of wage inequality in a globalized economy. Their research demonstrates that trade and market openness can exacerbate wage dispersion, as more productive firms pay higher wages while others lag behind. This highlights the importance of controlling for unemployment and market-driven wage gaps; social spending must be substantial enough to counteract these inherent structural pressures that push workers into the

lowest income percentiles. It is also important to note that some welfare programs may disincentivize work through providing alternate means of substantial income. This presents a problem of simultaneity and is partially accounted for in the lagged nature of the social spending variable.

The efficacy of specific spending types is further clarified by recent fiscal incidence studies. Lustig et al. (2014) demonstrate that in-kind transfers, specifically public spending on health and education, have a significantly greater impact on reducing inequality than cash transfers or direct taxation alone. Their analysis of Latin American economies reveals that while direct taxes are progressive, their impact on disposable income is often limited by low collection rates. In contrast, final income inequality is reduced most effectively when governments invest in human capital.

This supports the hypothesis that targeted public investment facilitates redistribution not just through immediate cash assistance, but by increasing the productive capacity of the workforce. Collectively, these studies suggest that an increase in social spending, particularly when channelled through robust democratic institutions and focused on human capital, should theoretically reduce the proportion of workers trapped in extreme relative poverty by mitigating both market-driven wage dispersion and the lack of access to essential public services.

Theoretical model

This paper will deduce if social spending has a positive effect on income inequality, a 3 year lag of social spending will be used to incorporate the fact that social spending can take a while to have an effect as well as reducing simultaneity between poverty rates

and social spending. Covariates such as Real GDP, Population, Education, Corruption(strength of institution), and Unemployment will be used to account for omitted variable bias as well as to account for growth rates in a country. A two way fixed effects regression will be done to account for country and year characteristics. An interaction between institutional integrity and social spending will produce an alternate hypothesis and offer insights into the conclusion of whether strong institutions increase the redistributive effect of social spending .

Therefore a relevant hypothesis is: An increase in social spending is positively associated with a decrease in the % of the population considered low income.

With an alternate hypothesis of: Better institutions increase the effect of social spending on reducing the amount of low income households.

Validity and Endogeneity

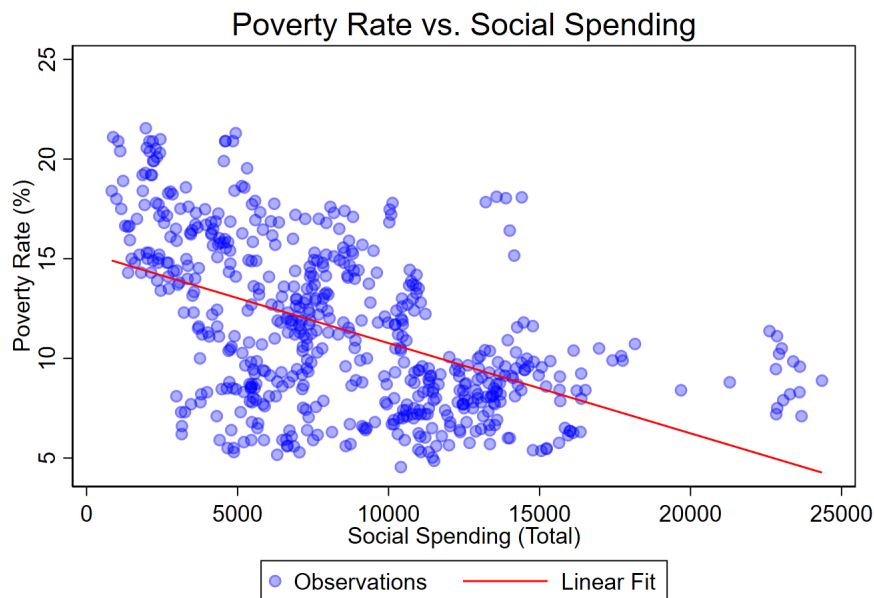
A major challenge in this study is endogeneity. Social spending is often spurred by an increased need for social spending and vice versa, the lag in the model aims to capture if lower spending in the past leads to higher poverty in the future and aims to remove this autocorrelation. Additionally, Omitted Variable Bias is a concern, as cultural attitudes toward social spending and whether it is ok to live on welfare as opposed to finding employment can have a significant effect on how useful social spending is, much of this is captured in state fixed effects however there is bound to be excess.

Expectations

While literature suggests progressivity and redistribution reduces income inequality, its impact on broad human development may be more complex. The study expects that the efficiency of converting public and mandatory private spending into social outcomes is mediated by institutional integrity.

Data and Methods

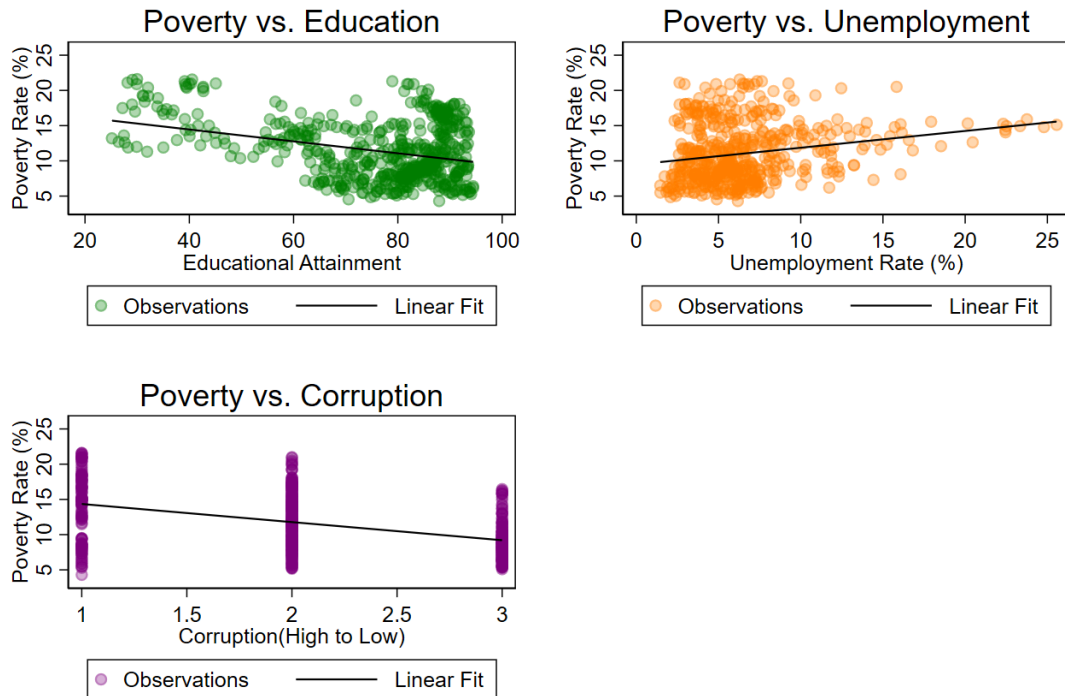
Relationship between dependent and independent



This graph shows that as social spending increases the poverty rate decreases, this is what our hypothesis expected. We can also see the closeness of the points to the best fit line, showing a large degree of variation.

Relationship between the dependent variable and control variables

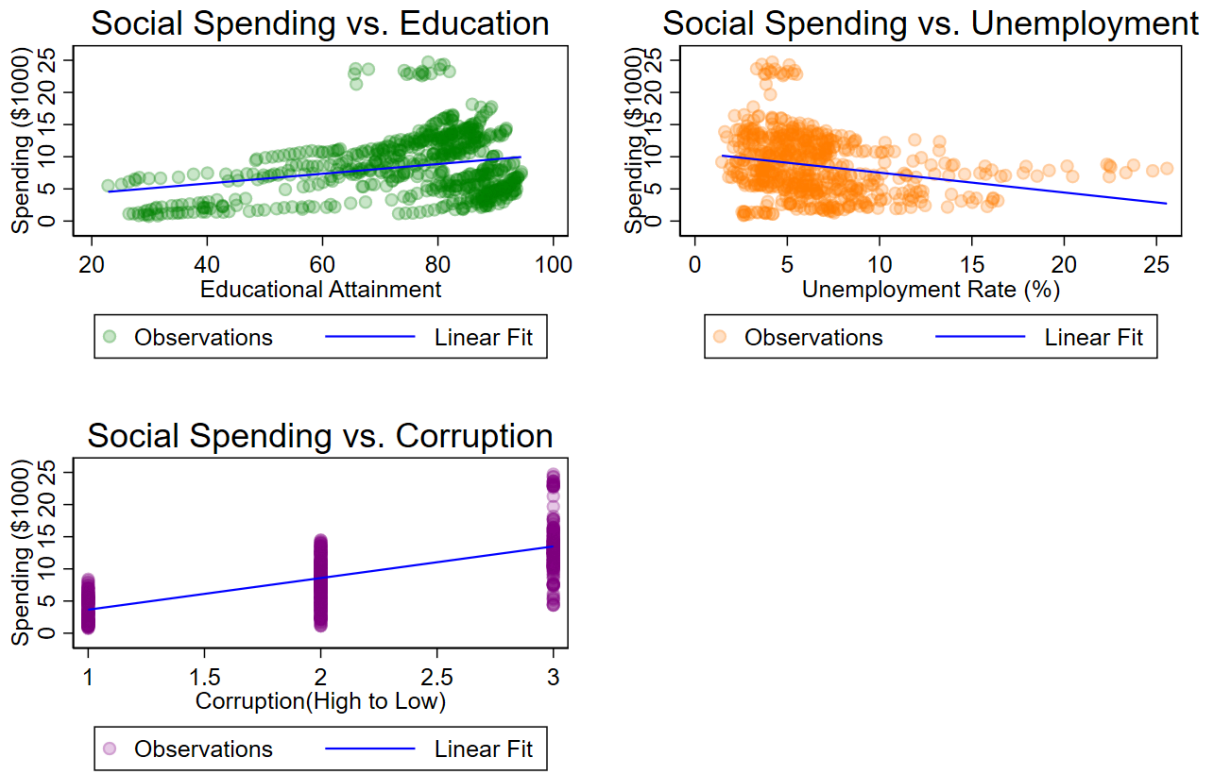
Poverty Dynamics



The education graph shows that there is weak negative correlation between upper secondary completion rates and the poverty rate. This is as expected, as ensuring a certain level of education across the population improves employability for those at the lower incomes in society. The relationship between unemployment and the poverty rate is positive, as expected, with higher levels of unemployment leading to higher poverty rates, however there is a large spread in the 2-10% unemployment range. The relationship between corruption and poverty is negative, as expected, as better institutions are likely to spend in more appropriate places and suffer less from the leaky bucket effect, therefore spending is better allocated and leads to lower poverty rates.

Relationship between the independent variable and control variables

Social Spending Determinants



These 3 graphs show what we would expect from the theory, as higher social spending figures are correlated with higher education attainment, lower unemployment (although this correlation isn't strong), and better institutions.

Summary of data analysis

I first address the relationship between my dependent variable, the poverty rate, and my primary independent variable, social spending. My empirical observations reveal a negative correlation, which directly supports my hypothesis that increased fiscal effort toward social welfare reduces the proportion of the population living below the poverty line. This relationship aligns with the Redistributive Hypothesis, suggesting that

state-led transfers effectively mitigate market-driven inequality. However, it is important to note that there is a significant degree of variation around the best-fit line. This variance is theoretically significant, as it suggests that the social return on spending is not uniform across all contexts, potentially pointing to differences in spending efficiency or the presence of the leaky bucket effect, where administrative costs or corruption diminish the impact of welfare capital.

When evaluating the Poverty Dynamics through control variables, I find results that confirm standard economic intuition. My analysis shows a positive relationship between unemployment and poverty, which strongly aligns with Keynesian labor market theory, where the loss of earned income serves as the primary driver of transitory poverty. The negative correlation between educational attainment and poverty aligns with Human Capital Theory which predicts that higher education levels should insulate individuals from poverty. Furthermore, the negative relationship between corruption and poverty confirms institutional economics frameworks; higher-quality, transparent institutions are more likely to implement distributive policies that successfully reach the marginalized rather than being captured by elites.

Finally, I assess the Social Spending Determinants to ensure the internal consistency of my model. My data confirms that higher social spending is positively correlated with higher educational attainment and lower corruption (better institutions). This is consistent with Wagner's Law, which posits that as societies become more complex and institutionally mature, the public demand for social insurance and welfare services grows. The strong correlation between institutional quality and spending justifies my decision to include an interaction term (Social spending x Corruption) in my regression,

as I must account for the fact that the effectiveness of a dollar spent is conditional upon the integrity of the state apparatus.

In conclusion, I find the dataset highly adequate for answering my research question. By utilizing a panel data structure with country and time fixed effects, I can control for unobserved heterogeneity that might otherwise bias my results. By utilising a 3-year-lagged spending measure I eliminate some inverse causality along with PPP converted measures of dollar amounts in order to capture the relative effect of spending.

Regression Equations

Without interaction

$$\text{Poverty}_{it} = \beta_0 + \beta_1 \text{SocialSpending}_{it-3} + \beta_2 \text{Corruption}_{it} + \beta_3 \text{GDP}_{it} + \beta_4 \text{Population}_{it} + \beta_5 \text{Education}_{it} + \beta_6 \text{Unemployment}_{it} + \alpha_i + \gamma_t + \mu_{it}$$

With interaction

$$\text{Poverty}_{it} = \beta_0 + \beta_1 \text{SocialSpending}_{it-3} + \beta_2 \text{Corruption}_{it} + \beta_3 (\text{SocialSpending}_{it-3} \times \text{Corruption}_{it}) + \beta_4 \text{GDP}_{it} + \beta_5 \text{Population}_{it} + \beta_6 \text{Education}_{it} + \beta_7 \text{Unemployment}_{it} + \alpha_i + \gamma_t + \mu_{it}$$

Where alpha and gamma are area and year fixed effects and μ is the error term.

In the second configuration the study also incorporates an interaction variable between institution quality and social spending.

The project employs a panel data analysis. By using Two-Way Fixed Effects, the model differences out permanent differences between countries and common shocks across

years (like worldwide economic shocks, disease or disasters) and hence only compares each country to itself over time, rather than comparing rich to poor countries. Standard errors are clustered at the country level to account for serial correlation within countries over time, ensuring the statistical significance of the results is not overstated.

Results

Table 1: The Impact of Social Spending and Corruption on Poverty Rates

[illegible]

Low Corruption X Spending							-3.220*** (0.843)	
Constant	44.845 *** (1.666)	43.268*** (2.102)	44.930*** (2.158)	40.363*** (2.511)	43.461*** (2.904)	28.471 (18.327)	85.518*** (16.127)	89.342*** (15.811)
Observations	578	576	567	497	481	481	481	481
R-squared	0.345	0.467	0.481	0.505	0.514	0.115	0.306	0.353
Country FE	No	No	No	No	No	Yes	Yes	Yes
Year FE	No	No	No	No	No	No	Yes	Yes
Number of country ID's						31	31	31

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.10

Models 1-5 use pooled OLS. Models 6-8 use within-group fixed effects.

The interaction term evaluates the moderating effect of institutions.

Discussion

The empirical results from Model 8, the preferred specification incorporating Two-Way Fixed Effects (TWFE) and an interaction term, provides a nuanced understanding of how fiscal policy interacts with institutional quality to shape poverty outcomes. By controlling for time invariant country characteristics and global economic shocks, this model isolates the conditional impact of social spending with high statistical rigor.

The most striking finding is the institutional multiplier effect. While social spending generally correlates with poverty reduction, its efficacy is not uniform across states. In high corruption environments, a 1% increase in social spending per capita yields a

0.01715% decrease in the poverty rate. Conversely, in low corruption environments, the same 1% increase results in a 0.04935% reduction.

Effectively, social spending in low corruption states is nearly three times more efficient than in states with high corruption. By looking at the average countries poverty rate, 11.4%, we can contextualize the effect of social spending and how it reduces the overall percentage of households considered low income in an economy and see that there is a significant difference which policy makers must address. From the results we can see that a 10% increase in spending per person reduces the number of low-income households by 5.3% in low-corruption contexts, compared to a reduction of 1.5% in highly corrupt ones. The interaction term for low-corruption countries is statistically significant at the 99% level ($P < 0.01$), providing robust evidence that the integrity of the state apparatus is a primary determinant of redistributive success. This aligns with the hypothesis that better institutions increase the effect of social spending on reducing the amount of low income households.

These findings validate and align with previous studies and hence this allows us to make conclusions as to why these distributive differences occur. These results confirm Okun's hypothesis that the bucket used to redistribute wealth is indeed prone to administrative and institutional leaks. In high corruption environments, these leaks, such as bureaucratic friction or fund capture by elites, are so pronounced that they nearly negate the impact of progressive fiscal policies. The results also support the idea that while democratic or high-spending capacity is important, the actual reduction of inequality depends on whether the political process is functional. My findings demonstrate that without institutional quality, the increased revenue available for public

programs fails to reach the intended marginalized demographics. We can also see that the statistically significant and negative correlation between education and poverty aligns with human capital theory, suggesting that education serves as a shield against economic volatility. Furthermore, the positive correlation between institutional maturity and spending levels aligns with Wagner's Law, where more complex societies demand more robust social insurance.

Policy Implications

The research suggests that for social spending to act as a sustainable pathway to prosperity rather than a temporary safety net, fiscal expansion must be accompanied by institutional reform. Therefore policymakers must view strengthening the state apparatus, such as reducing corruption and increasing transparency in government operations, as a prerequisite for effective poverty alleviation. Throwing money at the problem is an inefficient strategy if the redistributive mechanism is fundamentally eroded. From this we could also conclude that since the social return on a dollar spent is conditional on the administrative environment, international aid and domestic budget allocations should prioritize countries or regions that demonstrate higher institutional integrity to maximize the social multiplier and to have the greatest effect. These results also provide evidence to support the idea that social policy should lean towards in kind transfers, due to the significant impact of education on poverty reduction, as opposed to in cash transfers. This aligns with what historical literature suggests is a more optimal mix in increasing equality.

In conclusion, by proving that social spending is 3 times more effective at reducing poverty in low corruption countries we can say that while it is a powerful tool, it is not a

magic wand. Its power is either magnified or muffled by the quality of the institutions that wield it.

Variable Definitions + codebook

Type	Variable	Description	Calculated by	Link
Dependent	Income Inequality/ Poverty Rate	Percentage of the population whose disposable income is less than 50% of the median disposable income	Data until 2011 has slightly different definition of disposable income compared to post 2012 data	OECD
Independent	Social spending (lagged by 3 years)	Aggregate of all public and mandatory private social spending.	Natural log of dollars of social spending per person, PPP converted	OECD
Control	GDP	Real GDP of each country	GDP at constant prices in USD, PPP converted	OECD
Control	Population	Total population of each country	Raw value.	OECD
Control	Education	Percentage of population with at least upper secondary education (over 9 years of education)	% of 25-64 year olds with at least upper secondary education.	OECD
Control	Unemployment	Unemployment rate as a % of the labour force	% of labour force 25 years or older classed as unemployed	OECD
Control	Corruption	A categorical variable of low median and high degrees of corruption	Derived from 807 surveys covering 1,326,656 respondents (1995–2022). A	GDL

		(grouped by data trends, roughly 25:50:25 respectively)	combination of 2 sub indices: experienced petty corruption and perceived grand corruption.	
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References and Acknowledgements

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